

Prioritizing potential aquaculture

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Screenshot of GitHub README.md

(place holder)

Load required libraries

```
library(tidyverse)
library(sf)
library(tmap)
library(terra)
library(kableExtra)
```

Read data

```
# Read sea surface temperature data
sst_2008 <- rast(here::here('data', 'average_annual_sst_2008.tif'))
sst_2009 <- rast(here::here('data', 'average_annual_sst_2009.tif'))
sst_2010 <- rast(here::here('data', 'average_annual_sst_2010.tif'))
sst_2011 <- rast(here::here('data', 'average_annual_sst_2011.tif'))
sst_2012 <- rast(here::here('data', 'average_annual_sst_2012.tif'))

# Combine SST raster into a raster stack
sst_stack <- c(sst_2008, sst_2009, sst_2010, sst_2011, sst_2012,
              along = list(year = 2008:2012))

# Read Bathymetry data
depth <- rast(here::here('data', 'depth.tif'))

# Read Exclusive Economic Zones data
eez <- st_read(here::here('data', 'wc_regions_clean.shp'))
```

Prepare data

```
# Check CRS for all data set
st_crs(sst_stack)
st_crs(depth)
st_crs(eez)

# Convert the CRS of sst_stack to EPSG:4326
crs(sst_stack) <- "EPSG:4326"

# Check CRS of the dataset
if(st_crs(sst_stack) == st_crs(depth)){
  print("Coordinate reference systems match!")
} else {
  warning("Update coordinate reference systems to match!")
}
```

```
[1] "Coordinate reference systems match!"
```

Process data

Find the mean of SST stacks from 2008-2012

```
# Find the mean SST from 2008-2012 (e.g. create single raster of average SST)
sst_mean <- app(sst_stack, mean, na.rm = TRUE)

# Convert average SST from Kelvin to Celsius
sst_mean_celsius <- sst_mean - 273.15
```

Crop depth raster to match the extent of the SST raster

```
# Crop the ocean depth
depth_cropped <- crop(depth, ext(sst_mean_celsius))
```

Resample the depth data to match the resolution of the SST data using the nearest neighbor approach

```
# Resample the ocean depth data
depth_resample <- resample(depth_cropped, sst_mean_celsius, method = 'near')
```

Check that the ocean depth and SST match in resolution, extent and coordinate reference system - *Can the rasters be stacked?*

```
# Can the rasters be stacked?
if(all.equal(ext(depth_resample), ext(sst_mean_celsius)) == TRUE &&
   all.equal(res(depth_resample), res(sst_mean_celsius)) == TRUE &&
   crs(depth_resample) == crs(sst_mean_celsius)) {
  test_stack <- c(depth_resample, sst_mean_celsius)
  print('Rasters can be successfully stacked!')
} else {
  warning('Rasters cannot be stacked. Compatibility issue detected!')
}
```

```
[1] "Rasters can be successfully stacked!"
```

Find suitable locations for Oysters

Reclassify SST and ocean depth data into locations that are suitable for Oysters

```
# Reclassify SST into locations that are suitable for oysters
sst_reclass_oysters <- matrix(c(-Inf, 11, 0,
                               11, 30, 1,
                               30, Inf, 0), ncol = 3, byrow = TRUE)

sst_classify_oysters <- classify(sst_mean_celsius, sst_reclass_oysters)

depth_reclass_oysters <- matrix(c(-Inf, -70, 0,
                                  -70, 0, 1,
                                  0, Inf, 0), ncol = 3, byrow = TRUE)

depth_classify_oysters <- classify(depth_resample, depth_reclass_oysters)

# Find suitable locations
suitable_locations_oysters <- sst_classify_oysters * depth_classify_oysters
```

Determine the most suitable EEZ

Select suitable cells within West Coast EEZs

```
# Check and transform coordinate reference systems
if(st_crs(eez) == crs(sst_mean_celsius)) {
  print("Coordinate reference systems match!")
} else {
  warning("Updating coordinate reference systems to match!")
}
```

Warning: Updating coordinate reference systems to match!

```
# Transform economic zones data set to match raster CRS
eez <- st_transform(eez, crs(sst_mean_celsius))
```

```
# Convert the eeZ polygons to raster using the regions
eez_raster <- rasterize(eez, suitable_locations_oysters, field = "rgn_id")
```

Find area of grid cells

```
# Calculate the area in km square
area_km2_cell <- cellSize(suitable_locations_oysters, unit = "km")
```

Find the total suitable area within each EEZ

```
# Calculate the suitable area
suitable_area <- suitable_locations_oysters * area_km2_cell

# Calculate suitable area for each eeZ polygons
eez_suitable_area <- extract(suitable_area, eeZ, fun = "sum", na.rm = TRUE)

# Add the eeZ suitable area to the eeZ dataset
eez$suitable_area_km2 <- eeZ_suitable_area[, 2]

# Sort by suitable area (descending)
eez_rank <- eeZ[order(-eez$suitable_area_km2), ]

# Convert to data frame and select columns
eez_df <- as.data.frame(eez_rank)[, c("rgn", "suitable_area_km2")]

# Visualize the eeZ rank in table format
kable(eez_df,
      caption = "Suitable economic zones in West Coast for Oyster aquaculture",
      col.names = c("Region", "Suitable area (per km2)"),
      digits = 2,
      row.names = FALSE,
      format.args = list(big.mark = ",")) %>%
kable_styling(full_width = FALSE, font_size = 11)
```

Table 1: Suitable economic zones in West Coast for Oyster aquaculture

Region	Suitable area (per km ²)
Central California	4,923.15
Southern California	4,096.54
Washington	3,224.74
Oregon	1,533.09
Northern California	438.15

```
# Plot the suitability ranking for Oyster aquaculture
oyster_map <- tm_shape(eez_rank) +
  tm_polygons(fill = "suitable_area_km2",
              fill.scale = tm_scale(values = "plasma"),
              fill.legend = tm_legend(title = "Suitable area (per km2)",
                                     fontface = "bold",
```

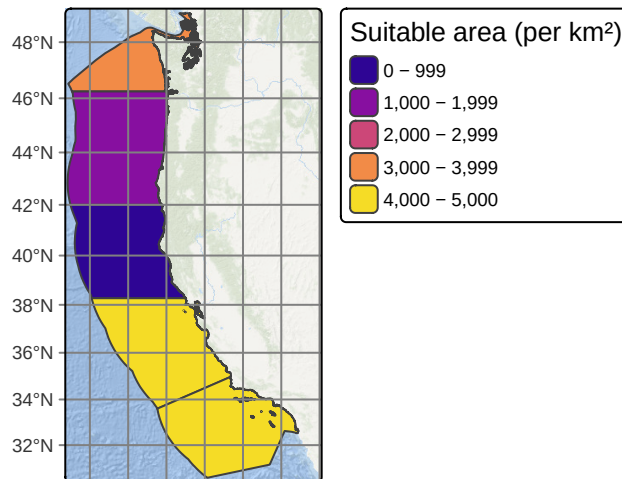
```

                                title.size = 0.8,
                                text.size = 0.6)) +
tm_title(text = "Suitability-based ranking of West Coast EEZs\n for Oyster aquaculture",
        size = 0.9, fontface = "bold") +
tm_graticules() +
tm_basemap(server = "Esri.OceanBasemap", alpha = 0.8) +
tm_layout(inner.margins = c(0.01, 0.01, 0.01, 0.1),
          component.autoscale = FALSE)

oyster_map

```

Suitability-based ranking of West Coast EEZs for Oyster aquaculture



Building a generalizable workflow

Here, we'll combine the above workflow above into a single function to ensure the process is clear and reproducible.

I will be determining which Exclusive Economic Zones (EEZ) on the West Coast of the US are best suited for developing marine aquaculture for **California mussel** (*Mytilus californianus*). The suitable sea surface temperature (SST) is usually 8.6 - 14.3 degree Celsius and depth values is 0 - 30 meters for this species.

```

# Function to find suitable aquaculture locations and create outputs
aquaculture_location <- function(species_name, min_temp, max_temp,
                                min_depth, max_depth)
{
  # Reclassify SST based on temperature range
  sst_reclass <- matrix(c(-Inf, min_temp, 0,
                          min_temp, max_temp, 1,
                          max_temp, Inf, 0),
                        ncol = 3, byrow = TRUE)
  sst_classify <- classify(sst_mean_celsius, sst_reclass)

  # Reclassify depth based on depth range
  depth_reclass <- matrix(c(-Inf, min_depth, 0,
                           min_depth, max_depth, 1,
                           max_depth, Inf, 0),
                         ncol = 3, byrow = TRUE)
  depth_classify <- classify(depth_resample, depth_reclass)

  # Find suitable locations
  suitable_locations <- sst_classify * depth_classify

  # Convert EEZ polygons to raster
  eezeraster <- rasterize(eez, suitable_locations, field = "rgn_id")

  # Calculate area in km2
  area_km2_cell <- cellSize(suitable_locations, unit = "km")

  # Calculate suitable area
  suitable_area <- suitable_locations * area_km2_cell

  # Calculate suitable area for each EEZ polygon
  eezsuitable_area <- extract(suitable_area, eezeraster, fun = "sum", na.rm = TRUE)

  # Add suitable area to EEZ dataset
  eezeraster$suitable_area_km2 <- eezsuitable_area[, 2]

  # Sort by suitable area (descending)
  eezerank <- eezeraster[order(-eezeraster$suitable_area_km2), ]

  # Convert to data frame and select columns
  eezerdf <- as.data.frame(eezerank)[, c("rgn", "suitable_area_km2")]

```

```

# Visualize the eez rank in table format
eez_table <- kable(eez_df,
                  caption = paste0("Suitable economic zones in West Coast for ", species_name),
                  col.names = c("Region", "Suitable area (per km2)"),
                  digits = 2,
                  row.names = FALSE,
                  format.args = list(big.mark = ",")) %>%
  kable_styling(full_width = FALSE, font_size = 11)

# Create suitability map
species_map <- tm_shape(eez_rank) +
  tm_polygons(fill = "suitable_area_km2",
             fill.scale = tm_scale(values = "plasma"),
             fill.legend = tm_legend(title = "Suitable area (per km2)",
                                     fontface = "bold",
                                     title.size = 0.8,
                                     text.size = 0.6)) +
  tm_title(text = paste0("Suitability-based ranking of West Coast EEZs\n for ",
                        species_name, " aquaculture"),
          size = 0.9, fontface = "bold") +
  tm_graticules() +
  tm_basemap(server = "Esri.OceanBasemap", alpha = 0.8) +
  tm_layout(inner.margins = c(0.01, 0.01, 0.01, 0.1),
            component.autoscale = FALSE)

# Return both table and map as a list
return(list(table = eez_table, map = species_map))
}

```

Apply function for species of choice

```

# Apply function for California Mussel
california_mussel <- aquaculture_location(species_name = "California Mussel",
    min_temp = 8.6,
    max_temp = 14.3,
    min_depth = -30,
    max_depth = 0)

# Display the suitability table for California Mussel
california_mussel$table

```


Table 2: Suitable economic zones in West Coast for California Mussel aquaculture

Region	Suitable area (per km ²)
Washington	3,603.10
Central California	1,843.34
Oregon	1,443.52
Northern California	1,155.22
Southern California	318.81

```
# Display the suitability map for California Mussel
california_mussel$map
```

Suitability-based ranking of West Coast EEZs for California Mussel aquaculture

